

# **Deep Learning Based Pneumonia Classification with Model Explainability Using Grad-CAM and LIME**

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## 1. Introduction

Pneumonia is a serious lung infection that affects many people around the world and is still one of the main reasons for illness and death, especially in young children and older adults. A chest X-ray is a common way to check for pneumonia, but looking at these images by hand takes a lot of time and can lead to different results depending on who is looking. Because of this, there has been growing interest in using deep learning systems to help doctors analyze medical images more quickly and consistently.

Even though these deep learning models work well at predicting pneumonia, they are often criticized for not being easy to understand. This lack of clarity is a big problem in areas like healthcare where decisions can be life-saving. XAI, is a set of methods that helps make these models more transparent by showing how they come to their conclusions. This project gives us the clarity using deep learning technique for detecting pneumonia and uses XAI techniques to make the model's decisions accurate and more trustworthy for humans .

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### 1.1 Dataset Description(Task 1)

The **Chest X-ray** Pneumonia dataset taken from Kaggle served as the source for the experiments. It is a dataset of radiographic images that were sorted into two classes: **NORMAL** and **PNEUMONIA**. The images were taken from children's hospital patients, and they are commonly used in the research field of medical all imaging.

The dataset is provided with predefined splits:

- **Training set**
- **Validation set**
- **Test set**

This predefined separation ensures fair evaluation and prevents data leakage between training and testing phases. All images were resized to a fixed resolution of 224×224 pixels to maintain consistency across models.

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### 1.2 Data Preprocessing(Task 1)

Before model training, several preprocessing steps were applied:

- Images were resized to 224×224 pixels.
- Pixel values were converted to tensors.
- Normalization was applied to scale pixel intensities, which helps stabilize training and improve convergence.
- Input images and target labels were clearly separated during data loading.

These preprocessing steps are standard practice in deep learning based image classification and ensure compatibility with convolutional neural network architectures.

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## 2. Model Evaluation and Comparison (Task 2)

The evaluation of models was carried out through accuracy, precision, recall, and F1-score metrics. Because of the class imbalance, F1-score was given primary consideration, as it has the capability of balancing false positives and false negatives.

### 2.1 Baseline CNN

A custom convolutional neural network was used as a baseline model. It is made from a sequence of convolutional and pooling layers that go one after another and then connect to fully connected layers.

### 2.2 VGG16 (Transfer Learning)

VGG16 is a deep convolutional network that has been pretrained on the ImageNet database. The method of transfer learning was employed by keeping the feature extraction layers frozen and training a classifier head specifically for pneumonia detection.

### 2.3 ResNet50 (Transfer Learning)

ResNet50 utilizes residual connections that significantly enhance the gradient flow and allow for the creation of deeper representations. A pretrained model of ResNet50 was modified for binary classification leading to this being a potential model for the medical image analysis.

**Table 1: Performance Comparison of Models**

| Model         | Accuracy | Precision | Recall | F1-score      |
|---------------|----------|-----------|--------|---------------|
| Baseline CNN  | 0.8173   | 0.7816    | 0.9821 | <b>0.8705</b> |
| VGG16 (TL)    | 0.7420   | 0.7078    | 1.0000 | 0.8289        |
| ResNet50 (TL) | 0.7821   | 0.7471    | 0.9846 | 0.8496        |

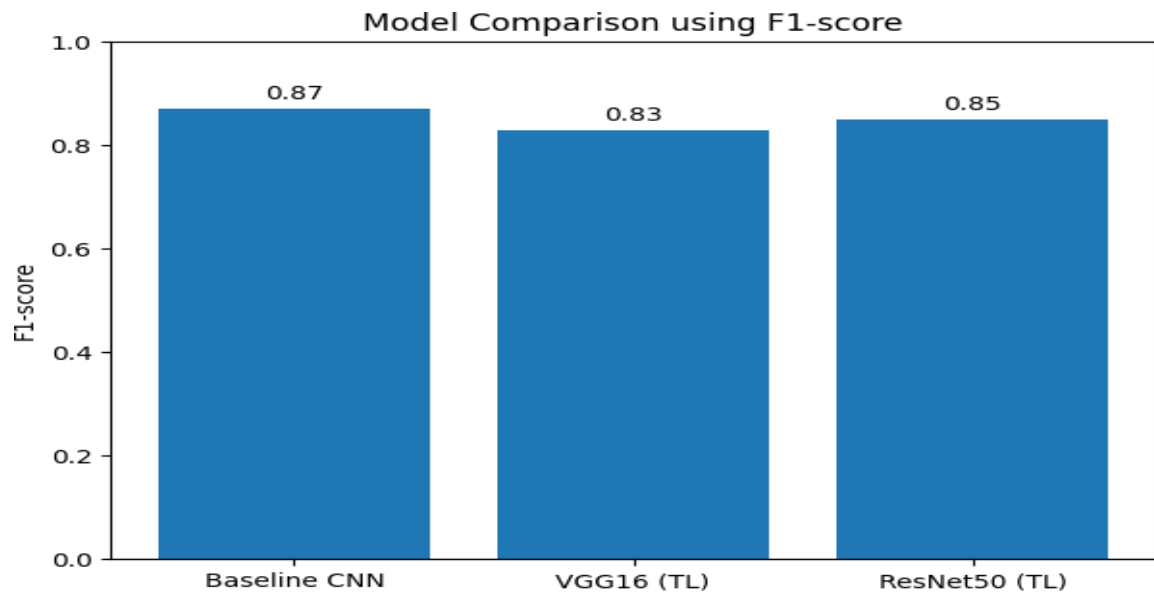


Figure 1: Model Comparison using F1-score

This illustration just exposes the F1-scores of the three assessed models. The baseline CNN attains the top F1-score, demonstrating the most favorable trade-off between precision and recall. Therefore, it is the model to be used in the explainability analysis

## 2.4 Confusion Matrix Analysis (Task 2)

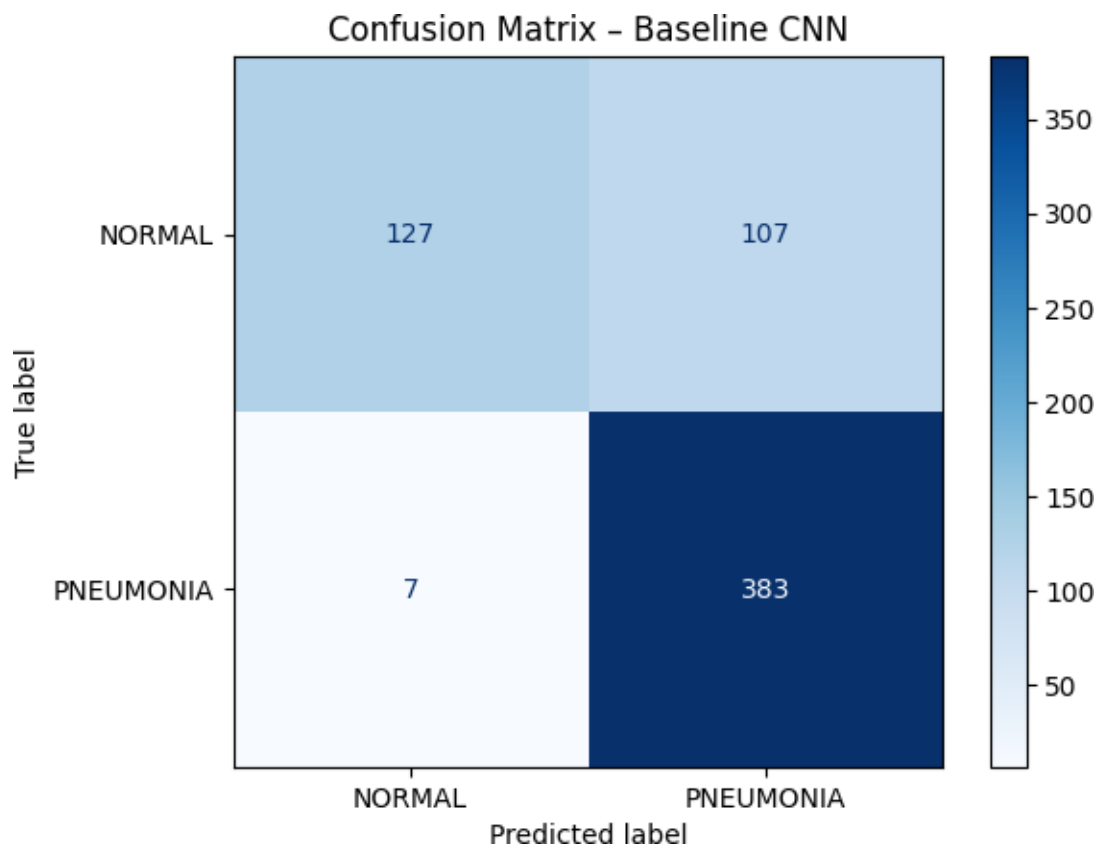


Figure 2: Confusion Matrix – Baseline CNN

The confusion matrix shows a high number of true positive pneumonia cases (383) and very few false negatives (7), confirming the model's strong recall. Although some false positives are present, this trade-off is acceptable in medical screening scenarios where missing a pneumonia case is more critical than raising a false alarm.

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### 3. Explainable AI Method 1: Grad-CAM (Tasks 3 & 4)

Grad-CAM was utilized on the baseline CNN, VGG16 and ResNet50 model for the purpose of showing which parts of the images were responsible for the predictions of pneumonia. Grad-CAM calculates maps of gradient-weighted activations from the last convolutional layer in order to point out the significant areas in the image.

#### 3.1 Grad-CAM Heatmap – Baseline CNN

The Grad-CAM heatmap produced here shows lung areas having more opacity, which is in line with the pathology of pneumonia. Moreover, the specific activation patterns imply that the model is dependent on the relevant lung features for diagnosis rather than on the noise in the background

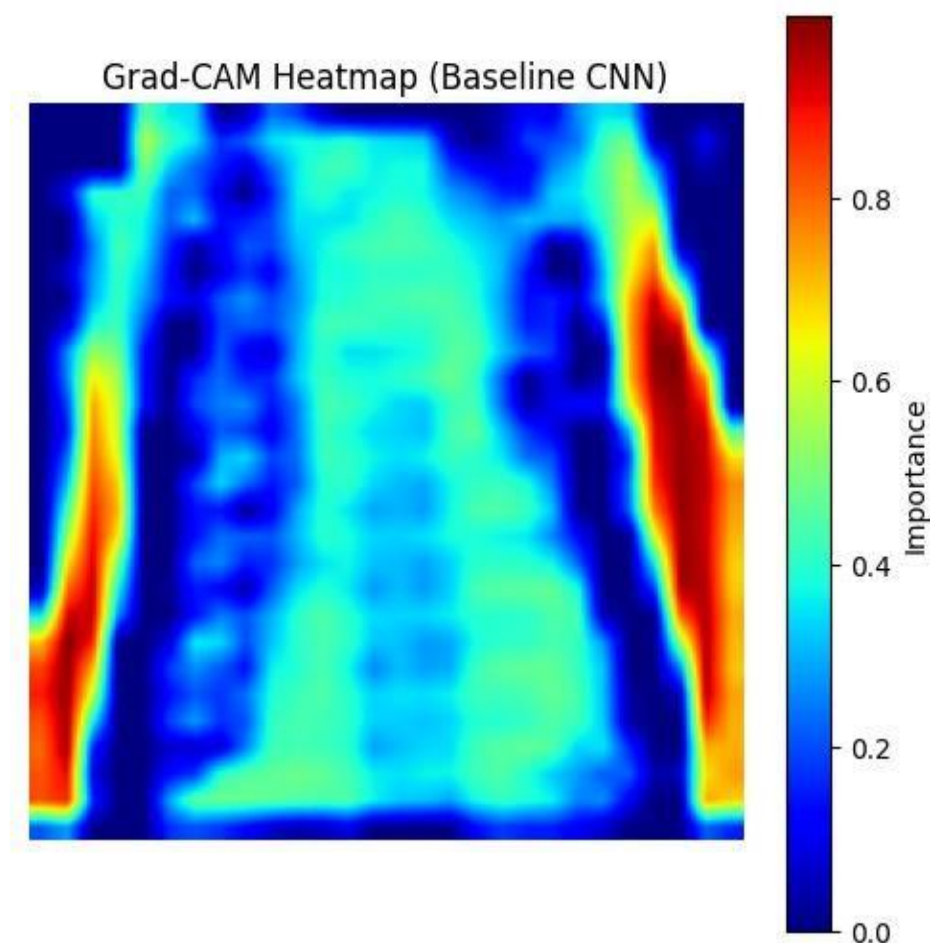


Figure 3: Grad-CAM Heatmap – Baseline CNN

### 3.2 Grad-CAM Heatmap – VGG16

The Grad-CAM output for VGG16 looks weak and homogenous as a result of the frozen convolutional layers in the transfer learning configuration. Such a situation prevents gradients from spreading and demonstrates the well known limitation that gradient-based XAI techniques suffer from when used with frozen feature extractors.

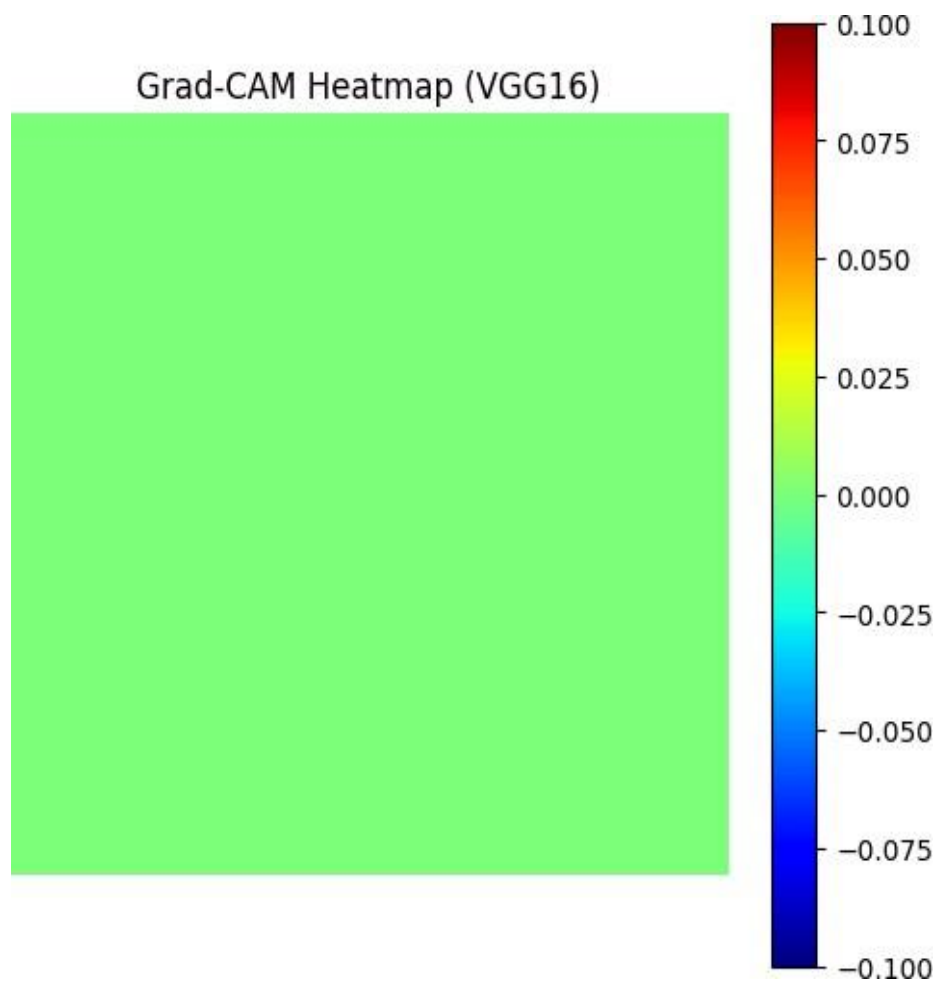


Figure 4: Grad-CAM Heatmap – VGG16

### 3.3 Grad-CAM Heatmap – ResNet50

ResNet50 gives rise to activation maps that are both structured and spatially meaningful, directing attention to the lung areas that are significant for pneumonia detection. The use of residual connections leads to better gradient flow, hence the explanations obtained are clearer compared to those produced by VGG16.

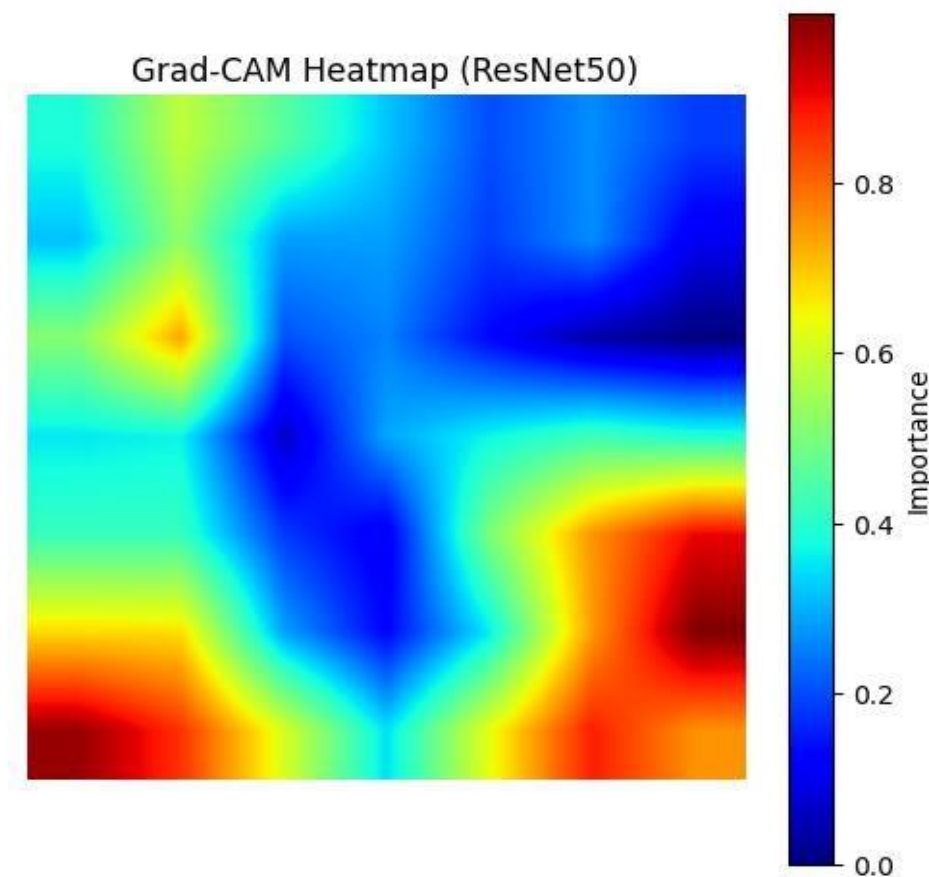


Figure 5: Grad-CAM Heatmap – ResNet50

#### Interpretation (Task 4)

The Grad-CAM heatmap shows which parts of the lungs the model focuses on when making its decision. These areas are where pneumonia often causes cloudiness or thickening in the lungs due to inflammation. The highlighted regions match the typical signs doctors look for in pneumonia. The model seems to pay more attention to the edges of the lungs, probably because it notices changes in texture and brightness that are linked to unhealthy lung tissue.

Grad-CAM results indicate that the baseline CNN and ResNet50 models base their predictions on medically meaningful lung regions. This alignment with domain knowledge improves trust in the model's predictions. However, Grad-CAM provides coarse spatial explanations and may overlook fine-grained features, limiting pixel-level interpretability

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#### 4. Explainable AI Method 2: LIME (Tasks 5 & 6)

Local Interpretable Model-agnostic Explanations (LIME) was selected as a second XAI method to provide an alternative perspective. LIME explains individual predictions by perturbing local regions of the input image and learning a simple surrogate model that approximates the classifier's behavior. LIME (Local Interpretable Model-agnostic Explanations) was used to provide instance-level explanations by perturbing superpixels and observing changes in model predictions.

LIME Explanation (Baseline CNN)

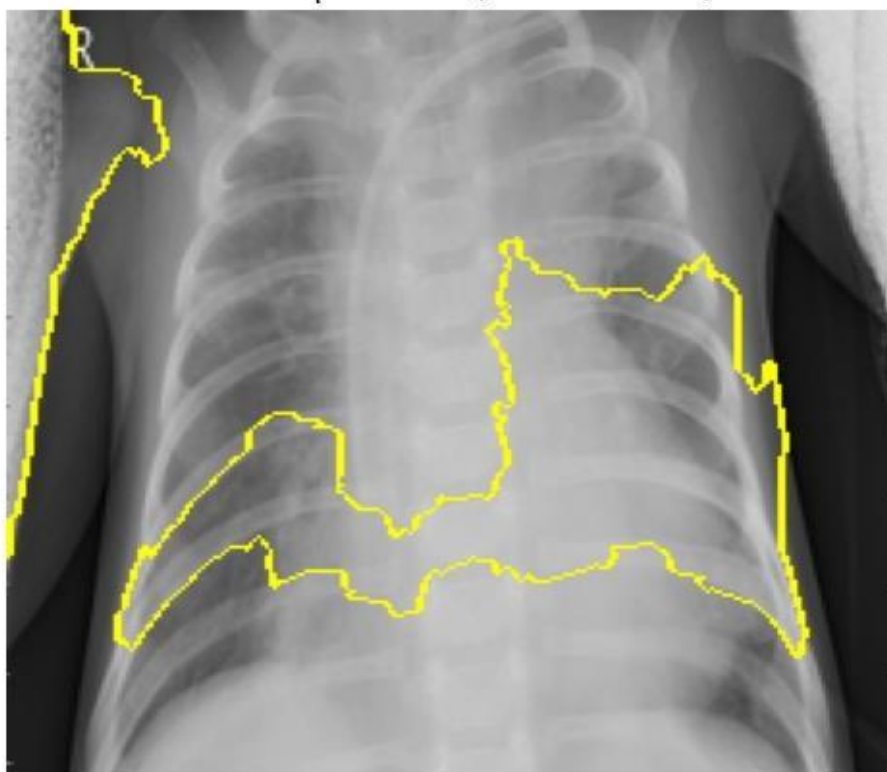


Figure 6: LIME Explanation – Baseline CNN

LIME Explanation (VGG16)

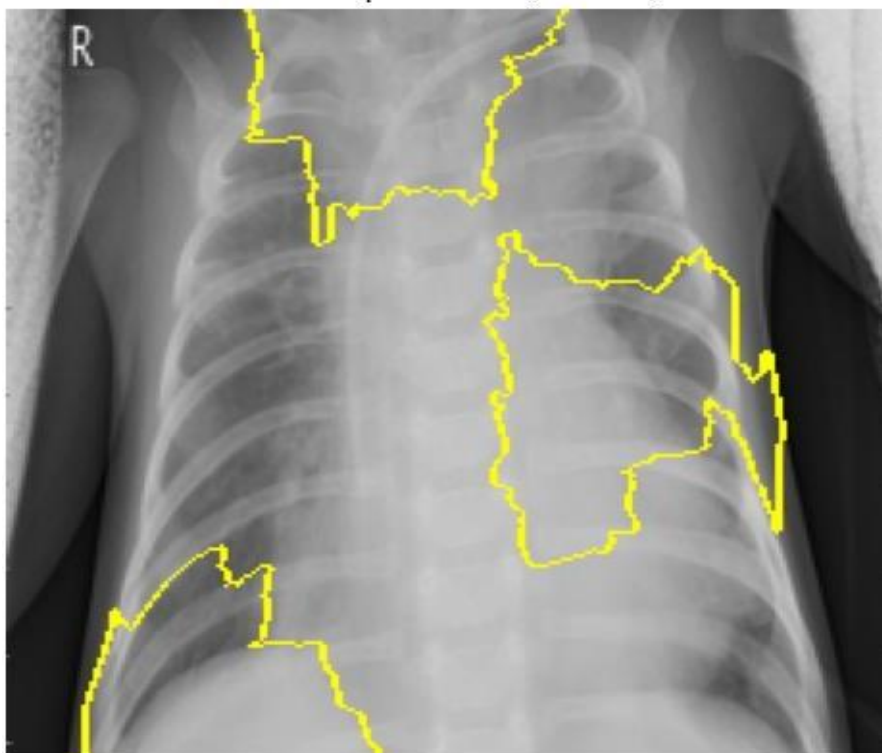


Figure 7: LIME Explanation – VGG16



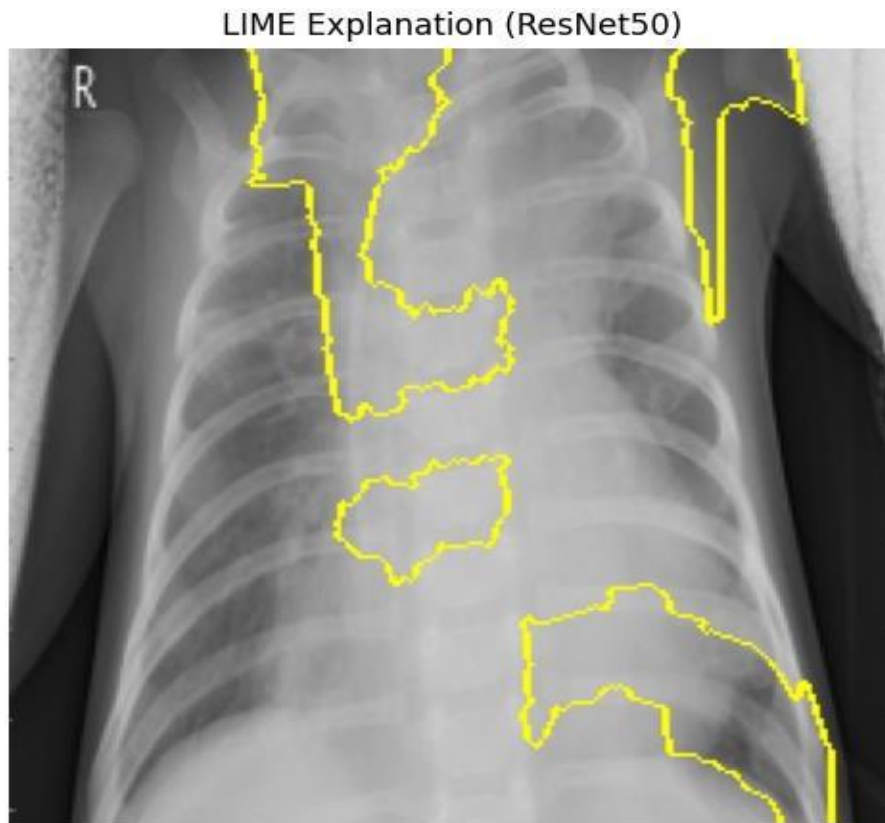


Figure 8: LIME Explanation – ResNet50

**Description:** LIME focuses on superpixel regions that are localized in the lung area and have a positive impact on the pneumonia predictions. It is different from Grad-CAM in that it does not use gradients, thus it can still influence the prediction even if the convolutional layers are frozen, as it happens with VGG16.

### Comparison of Grad-CAM and LIME (Task 6)

While explaining LIME it shows that the model's prediction is based on certain areas in the lungs, which are made up of small regions called superpixels. This means the model is trying to focus on important parts of the body rather than just random details present on the background

Grad-CAM explains the model's decision by showing how different parts of the image activate across the whole image, using patterns from the layers in the network. LIME, on the other hand, looks at smaller parts present in the image and it tests how changing those parts will affect the model's decision. Grad-CAM creates smooth, clear images that match real body structures, while LIME creates detailed but sometimes broken-up areas. Together, these two methods give a more complete and trustworthy explanation of how the model works in that environment.

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## 5. Ethical Considerations and Limitations

Deep learning models are practiced in the medical imaging area, but there are still ethical issues to resolve first. False negatives could slow down the treatment process and false positives could

cause unnecessary anxiety or even treatments. On top of that, the dataset could be biased due to patients' age, sex, or race, or imaging conditions, thus impacting generalization. Translucent AI procedures have the capacity to lessen the impact of these risks by making the process more transparent, but still the uncertainty remains. As long as clinical AI systems are in place the presence of humans is mandatory

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## **6. Conclusion**

The baseline CNN from this research is the one that reached the highest F1-score and also gave the most comprehensible explanations among the tested models. Grad-CAM and LIME together assert that the model uses lung characteristics that are clinically significant for pneumonia diagnosis. It is the joining of AI with explainability that is the main factor in confirming the accuracy of deep learning models in medical imaging and in facilitating their responsible use.

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## **Appendix A: Code Availability**

The complete implementation for this assignment is provided in a Jupyter Notebook. It covers dataset loading and preprocessing, training of three models (Baseline CNN, VGG16 with transfer learning, and ResNet50 with transfer learning), model evaluation using accuracy, precision, recall, F1-score, and confusion matrices, as well as explainable AI analysis using Grad-CAM and LIME. All experiments were conducted in Python using Google Colab to ensure reproducibility.

The notebook is available at the following link:

<https://colab.research.google.com/drive/1J-oCdqcfJzkDjc-6vnBermkmMCJVGRgh?usp=sharing>